Provenance and covariance sensitivity in a summary-level comparison of sound-horizon-free Hubble-constant measurements

Ayoub Issa
Independent Researcher, Morocco
ORCID: 0009-0006-9500-1883

Manuscript version 1.0 — October 26, 2026

Abstract

Published compilations of Hubble-constant (H_0) estimates combine results that may share data, calibration anchors, likelihood components or model assumptions, yet the joint covariance is rarely available. We audit a recent 88-entry compilation of 30 distance-ladder and 58 other sound-horizon-free estimates at the summary level. Using the values and error symmetrisation of the pinned public analysis script, we reproduce inverse-variance means of 72.7325 ± 0.3852 and $69.3722 \pm 0.3449\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (nominal contrast $Z = 6.499$ under independence) and, for the source paper's illustrative within-group equicorrelations $(\rho_L, \rho_R) = (0.40, 0.15)$, generalised-least-squares (GLS) means of 72.3020 ± 0.5713 and $69.5205 \pm 0.4204\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ ($Z = 3.922$ with zero cross-group covariance). We then develop the dependence structure analytically: (i) an $O(n)$ closed form for equicorrelated GLS, with an exact threshold for the sign of each weight; (ii) the design-effect ceiling $n_{\mathrm{eff}} \to 1/\rho$; (iii) the exact positive-semidefinite limit on a cross-group correlation, here $|\rho_{LR}| \leq 0.263$; (iv) the monotone response of Z to the correlation κ between the two pooled means, giving $2.80 \leq Z \leq 18.4$ for the GLS contrast as κ ranges over $[-1, 1]$; and (v) a sharp worst-case variance under an entrywise correlation cap. We also record one provenance mismatch: the tabulated value 69.37 ± 0.65 is attributed to a 2024 DESI Year-1 preprint that reports a different number, whereas a 2025 presentation reports the matching value from DESI DR2. We additionally reconstruct the 88-row table from the source paper's appendix (ladder group reproduced exactly; comparison group to within $0.004\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$) and find that the equicorrelated model reproduces the reported 16 and 38 negative weights, that the feasible cross-group correlation confines Z to 3.75–4.12, and that random LKJ correlation matrices leave the independence value essentially unchanged ($Z \in [5.7, 7.6]$ at 95%), so a reduction to ≈ 3.9 requires sign-coherent shared systematics rather than generic dependence. All statistics are conditional summary-level quantities. The analysis does not show that the tension has disappeared, that its true significance takes any particular value, or that any measurement should be excluded. A companion C++17/Python library implements every result.

Keywords: Hubble constant; Hubble tension; covariance; meta-analysis; generalised least squares; reproducibility.

1 Introduction

The Hubble tension is the disagreement between determinations of the present expansion rate from different methods. Pantos and Perivolaropoulos [1] assembled 88 measurements that do not use the CMB sound horizon, separated 30 distance-ladder entries from 58 others, and reported a contrast of about 6.5σ under independence that falls to a minimum of 3.9σ once illustrative correlations are applied. A table of published summaries, however, cannot by itself guarantee that its entries target identical parameters or have a known joint covariance. This paper asks two narrow questions: *how much does the nominal significance depend on what is assumed about dependence*, and *which entries have source metadata that cannot be traced unambiguously.*

Claims ledger. To keep the scope explicit we separate what the audit establishes from what it does not.

Identified for scrutiny	Not shown here
The source/version mismatch of one tabulated estimate (69.37 ± 0.65), and the degree of dependence among estimates that rest on shared inputs.	That the tension has disappeared; that its true significance equals any particular value; that any specific measurement must be excluded. The full covariance matrix required for such statements is not available.

Contributions. (1) An independent reconstruction of the summary-level baseline (Section 4). (2) Exact, checkable results on equicorrelated GLS, its negative weights, its information ceiling, and the admissible range of a cross-group correlation (Section 3). (3) A sharp bound on the contrast variance under documented correlation caps (Section 3.6). (4) A targeted provenance finding (Section 6). (5) A reference implementation in C++17 with a Python mirror and a test suite (Section 7). The paper proposes no new cosmological model and no new measurement of H_0.

2 Data and estimators

Source snapshot. The table was taken from the public repository accompanying [1] at the immutable commit and file blob given in [2]. The script encodes 30 distance-ladder estimates (L) and 58 comparison estimates (R). We analyse summary values y_i with symmetrised errors s_i (root-mean-square of the upper and lower errors, as upstream); primary likelihoods are not refitted.

Independence baseline. With $a_i = s_i^{-2}$, $w_i = a_i / \sum_j a_j$, the pooled mean and error are $\widehat{H} = \sum_i w_i y_i$ and $\mathrm{SE} = (\sum_i a_i)^{-1/2}$.

Equicorrelated GLS. For a group of n estimates let $C = SRS$ with $S = \mathrm{diag}(s_i)$ and $R_\rho = (1-\rho)I + \rho\,\mathbf{1}\mathbf{1}^\top$. The GLS (Aitken [3]) mean and variance are

$$\widehat{H}_{\mathrm{GLS}} = \frac{\mathbf{1}^\top C^{-1} y}{\mathbf{1}^\top C^{-1}\mathbf{1}}, \qquad \mathrm{Var}(\widehat{H}_{\mathrm{GLS}}) = \left(\mathbf{1}^\top C^{-1}\mathbf{1}\right)^{-1}. \tag{1}$$

Following [1] we use $\rho_L = 0.40$, $\rho_R = 0.15$. These are assumptions, not estimates.

Contrast. $\widehat{\Delta} = \widehat{H}_L - \widehat{H}_R$ and $\mathrm{Var}(\widehat{\Delta}) = V_L + V_R - 2\,\mathrm{Cov}(\widehat{H}_L, \widehat{H}_R)$. The packaged calculations set the covariance to zero; $Z = |\widehat{\Delta}|/\mathrm{SE}(\widehat{\Delta})$ is a *nominal Gaussian-equivalent statistic*, not a calibrated p-value.

3 Mathematical results

All statements below are proved in the text or in `docs/theory.md`, and each is verified numerically by the test suite (Section 7).

3.1 Closed-form equicorrelated GLS and the sign of its weights

Theorem 1 (Closed form)**.** *Let $n \geq 2$, $\rho \in (-\frac{1}{n-1}, 1)$, $a_i = 1/s_i$, $A = \sum_i a_i$ and $c = \rho/(1+(n-1)\rho)$. Then*

$$(C^{-1}\mathbf{1})_i = \frac{a_i\,(a_i - cA)}{1-\rho}, \qquad N \equiv \mathbf{1}^\top C^{-1}\mathbf{1} = \frac{1}{1-\rho}\Big(\sum_i a_i^2 - cA^2\Big) > 0, \tag{2}$$

and the GLS weights are $w_i = (C^{-1}\mathbf{1})_i / N$, with $\mathrm{Var}(\widehat{H}_{\mathrm{GLS}}) = 1/N$.

Proof. By Sherman–Morrison [4], $R_\rho^{-1} = (1-\rho)^{-1}(I - c\,\mathbf{1}\mathbf{1}^\top)$, and $C^{-1} = S^{-1}R_\rho^{-1}S^{-1}$. Hence $C^{-1}\mathbf{1} = S^{-1}R_\rho^{-1}a = (1-\rho)^{-1}S^{-1}(a - cA\mathbf{1})$, which is the stated vector. Positivity of N: for $\rho > 0$, Cauchy–Schwarz gives $A^2 \le n\sum a_i^2$ and $cn < 1$, so $cA^2 < \sum a_i^2$; for $\rho \le 0$, $c \le 0$ and the claim is immediate. □

This replaces an $O(n^3)$ solve by $O(n)$ arithmetic and exposes the structure of the weights.

Corollary 1 (Negative weights)**.** *For $\rho > 0$ the weight of row i is negative if and only if*

$$s_i > s^* \equiv \frac{1 + (n-1)\rho}{\rho \sum_j s_j^{-1}}. \tag{3}$$

In particular no weight is negative when all s_i are equal.

Proof. $w_i < 0 \iff a_i < cA$ since $N > 0$ and $a_i > 0$; rewrite with $s_i = 1/a_i$. □

Negative GLS weights are therefore not a numerical accident: they are assigned exactly to the least precise rows, and their number is $\#\{i : s_i > s^*\}$. For the packaged scenario the upstream results report 16 of 30 and 38 of 58 negative weights; the command `h0audit summary` prints s^* and the count so the correspondence can be checked on the pinned table.

3.2 Design effect and an information ceiling

Proposition 1. *If all $s_i = s$, then $\mathrm{Var}(\widehat{H}_{\mathrm{GLS}}) = s^2\,[1 + (n-1)\rho]/n$, so the effective number of independent estimates is $n_{\mathrm{eff}} = n/[1 + (n-1)\rho]$ (Kish's design effect [5]), and $n_{\mathrm{eff}} \uparrow 1/\rho$ as $n \to \infty$.*

In this equal-error idealisation $(n, \rho) = (30, 0.40)$ gives $n_{\mathrm{eff}} = 2.38$ and $(58, 0.15)$ gives 6.07, against ceilings of 2.5 and 6.67 (Figure 4): under the assumed correlations, adding more rows barely tightens the pooled error. *This is a property of the assumed ρ, not a measurement of information duplication.* Likewise the weight share of three DESI DR2-input rows (Section 6) must not be read as an independent-information fraction.

3.3 Interpreting ρ: a shared-systematic factor

R_ρ is the correlation matrix of $\varepsilon_i = s_i(\sqrt{\rho}\,\eta + \sqrt{1-\rho}\,\xi_i)$ with $\eta, \xi_i \sim \mathcal{N}(0,1)$ independent. Equivalently

$$V = Z\,\Omega\,Z^\top + D, \qquad Z = s,\ \Omega = \rho,\ D = (1-\rho)S^2, \tag{4}$$

so ρ is the fraction of every row's variance that is a single common systematic. A documented dependency model would instead take Z to be a 0/1 design matrix of shared datasets, calibrations or pipelines and $\Omega \succeq 0$ the covariance of those shared effects, with Ω constrained by likelihoods or covariance products and *never* tuned to reach a preferred significance.

3.4 How large can a cross-group correlation be?

Suppose both groups are equicorrelated and every pair $(i \in L, j \in R)$ has correlation ρ_{LR}.

Theorem 2 (Exact PSD limit)**.** *The joint correlation matrix is positive semidefinite if and only if $-\frac{1}{n_g - 1} \le \rho_g \le 1$ for $g = L, R$ and*

$$\rho_{LR}^2 \le \frac{[1 + (n_L - 1)\rho_L]\,[1 + (n_R - 1)\rho_R]}{n_L n_R}. \tag{5}$$

Proof. Vectors orthogonal to $\mathbf{1}$ within one block are eigenvectors with eigenvalues $1-\rho_g \geq 0$. On the invariant plane spanned by $(\mathbf{1}_L/\sqrt{n_L}, 0)$ and $(0, \mathbf{1}_R/\sqrt{n_R})$ the matrix acts as

$$\begin{pmatrix} 1+(n_L-1)\rho_L & \rho_{LR}\sqrt{n_L n_R} \\ \rho_{LR}\sqrt{n_L n_R} & 1+(n_R-1)\rho_R \end{pmatrix},$$

which is PSD iff its determinant is non-negative (its diagonal entries are positive). □

For $(n_L, \rho_L, n_R, \rho_R) = (30, 0.40, 58, 0.15)$ the bound is $|\rho_{LR}| \leq 0.2630$. For linear estimators with weights w_L, w_R the pooled-mean covariance is $\mathrm{Cov}(\widehat{H}_L, \widehat{H}_R) = \rho_{LR}\,(w_L \cdot s_L)(w_R \cdot s_R)$. The library evaluates Z across the feasible interval on the real table.

3.5 Response of the significance to cross-group dependence

Let $\kappa = \mathrm{Corr}(\widehat{H}_L, \widehat{H}_R) \in [-1, 1]$.

Proposition 2. *With* $\mathrm{SE}_g = \sqrt{V_g}$,

$$Z(\kappa) = \frac{|\widehat{\Delta}|}{\sqrt{\mathrm{SE}_L^2 + \mathrm{SE}_R^2 - 2\kappa\,\mathrm{SE}_L \mathrm{SE}_R}} \tag{6}$$

is strictly increasing in κ*, with envelope* $|\widehat{\Delta}|/(\mathrm{SE}_L + \mathrm{SE}_R) \leq Z \leq |\widehat{\Delta}|/|\mathrm{SE}_L - \mathrm{SE}_R|$.

Thus the zero-covariance assumption of the reproduction is *not* a conservative simplification in general: positive cross-group error correlation (for example from a shared supernova sample) shrinks the variance of the difference and *raises* Z, while negative correlation lowers it. For the GLS means ($\widehat{\Delta} = 2.7815$, $\mathrm{SE}_L = 0.5713$, $\mathrm{SE}_R = 0.4204$) the envelope is $2.80 \leq Z \leq 18.4$, with $Z = 3.92$ at $\kappa = 0$ and $Z \geq 3.92$ for every $\kappa \geq 0$ (Figure 2). The lower end requires perfect anticorrelation of the two pooled means and is shown for completeness; inside the block-equicorrelated model of Theorem 2 the attainable range of κ is much narrower (Section 5). These values condition on the two within-group models being correct; they are not a probability statement about H_0.

3.6 Sharp worst case under an entrywise correlation cap

Let $q = (w_L \circ s_L,\ -w_R \circ s_R)$ for fixed estimator weights, so $\mathrm{Var}(\widehat{\Delta}) = q^\top R q$.

Proposition 3 (Sharp cap bound)**.** *Over all correlation matrices with* $|R_{ij}| \leq r$ $(i \neq j)$, $0 \leq r \leq 1$,

$$\max_R\, q^\top R q = (1-r)\sum_i q_i^2 + r\Big(\sum_i |q_i|\Big)^2, \tag{7}$$

attained by $R = (1-r)I + r\,\sigma\sigma^\top$ *with* $\sigma_i = \mathrm{sign}\, q_i$.

Proof. $q^\top R q = \sum q_i^2 + \sum_{i\neq j} q_i q_j R_{ij} \leq \sum q_i^2 + r\sum_{i\neq j} |q_i q_j|$, which equals the right-hand side. The stated R has unit diagonal, entries of modulus r, and is PSD for $r \in [0,1]$ (a convex combination of I and a rank-one PSD matrix). □

For $r = 1$ the bound is $(\sum |q_i|)^2$, the ℓ_1 worst case. Using the packaged inverse-variance weights, the corresponding minimum nominal significance $Z_{\min}(r)$ is

r	0	0.01	0.025	0.05	0.10	0.20	0.50	1.0
$Z_{\min}$	6.50	5.33	4.36	3.51	2.68	1.98	1.29	0.92

(Figure 3; values from `results/legacy_v1/hubble_tension_robust_bounds.csv`). Without documented caps the bound is uninformative at large r, which is the point: any statement about significance must name the correlation set it is conditional on.

Table 1: Reconstructed summary estimates (units $\mathrm{km\,s^{-1}\,Mpc^{-1}}$; zero cross-group covariance).

Estimator	Ladder	Comparison	$\widehat{\Delta}$	SE($\widehat{\Delta}$)	Nominal Z
IVW, independence	72.7325 ± 0.3852	69.3722 ± 0.3449	3.3602	0.5171	6.499
GLS, ρ_L=0.40, ρ_R=0.15	72.3020 ± 0.5713	69.5205 ± 0.4204	2.7815	0.7093	3.922

Table 2: Influence diagnostics (six scenarios). GLS rows keep $\rho_R = 0.15$ unchanged after exclusions, so they are diagnostics, not alternative best estimates.

Scenario	N_L/N_R	$\widehat{\Delta}$	SE($\widehat{\Delta}$)	Z
All rows; IVW	30/58	3.3602	0.5171	6.499
Excl. 69.37 ± 0.65; IVW	30/57	3.3594	0.5603	5.995
Excl. three DESI DR2-input rows; IVW	30/55	4.1933	0.6391	6.561
All rows; GLS	30/58	2.7815	0.7093	3.922
Excl. 69.37 ± 0.65; GLS	30/57	2.7177	0.7755	3.505
Excl. three DESI DR2-input rows; GLS	30/55	4.1665	0.9147	4.555

3.7 Random correlation structures: why generic dependence barely moves Z

Let $R \sim \mathrm{LKJ}(\eta)$ in dimension d (uniform over correlation matrices for $\eta = 1$; concentrated at I as η grows) [6]. Each off-diagonal entry has law $2\,\mathrm{Beta}(\eta{-}1{+}\frac{d}{2}, \eta{-}1{+}\frac{d}{2}) - 1$, hence mean 0 and variance $1/(d + 2\eta - 1)$.

Proposition 4 (Moments of the contrast variance)**.** *For fixed $q \in \mathbb{R}^d$, $\mathbb{E}[q^\top R q] = \sum_i q_i^2$ and*

$$\mathrm{Var}\big(q^\top R q\big) = \frac{2\big[(\sum_i q_i^2)^2 - \sum_i q_i^4\big]}{d + 2\eta - 1}. \tag{8}$$

Proof. LKJ is invariant under $R \mapsto DRD$ for D diagonal with entries ± 1, so $\mathbb{E}[R_{ij}R_{kl}] = 0$ unless $\{i,j\} = \{k,l\}$. Write $q^\top R q = \sum q_i^2 + 2\sum_{i<j} q_i q_j R_{ij}$ and sum the variances $4q_i^2 q_j^2/(d + 2\eta - 1)$ over pairs. □

Since $\sum_{i<j} q_i^2 q_j^2 = \frac{1}{2}[(\sum q_i^2)^2 - \sum q_i^4]$, the relative standard deviation of $\mathrm{Var}(\widehat{\Delta})$ is at most $\sqrt{2/(d + 2\eta - 1)}$, i.e. about 15% for $d = 88$, $\eta = 2$, and $Z \propto (q^\top R q)^{-1/2}$ moves by about half that. Unstructured dependence therefore *averages out*; a large reduction of Z needs correlations whose signs are aligned with q, exactly what a common-systematic factor (Section 3) produces.

3.8 Identifiability of the contrast

Write $y_i = H_* + \alpha_{g(i)} + \varepsilon_i$. The design matrix has rank two for three parameters $(H_*, \alpha_L, \alpha_R)$, and a linear functional $c^\top\theta$ is estimable iff $c^\top$ lies in the row space of the design. The contrast $\alpha_L - \alpha_R$ is estimable; H_* and the individual offsets are not without an external anchor or a stated constraint. When the cross-group covariance is zero the best linear unbiased estimator of the contrast is the difference of the group GLS means; with non-zero cross-covariance the joint Aitken estimator is the BLUE and the difference of separate group means is not, so the κ-analysis above describes that particular estimator.

4 Results on the packaged summaries

Tables 1 and 2 and Figure 1 reproduce the packaged computation. Two features deserve emphasis. First, the GLS contrast is not a property of the data alone: with the same (ρ_L, ρ_R) the estimator matters (Table 3). Unrestricted GLS, which permits negative weights, gives $Z = 3.92$; non-negative minimum-variance weights give 3.32; and fixed inverse-variance weights with the *same* covariance

Table 3: Estimator dependence at fixed $(\rho_L, \rho_R) = (0.40, 0.15)$, zero cross-group covariance.

Estimator	Ladder	Comparison	Nominal Z
Unrestricted GLS	72.302 ± 0.571	69.521 ± 0.420	3.92
Non-negative minimum-variance	72.830 ± 0.820	69.641 ± 0.499	3.32
Fixed IVW weights, covariance propagated	72.732 ± 1.204	69.372 ± 0.766	2.35

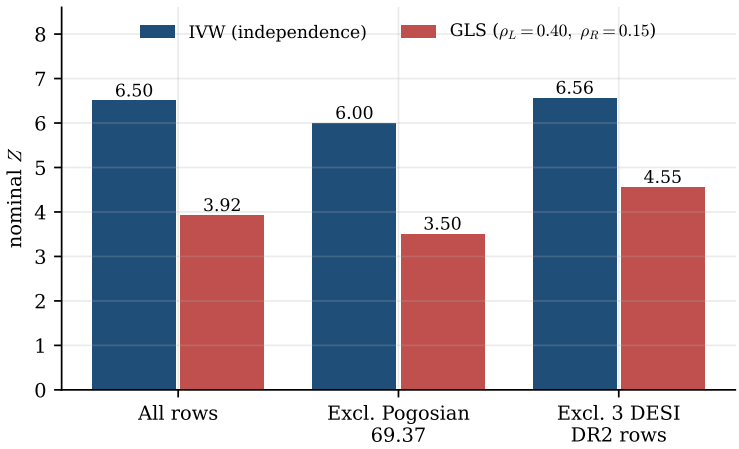


Figure 1: Nominal Z across the six scenarios of Table 2; conditional on the stated estimator and covariance assumptions.

propagated give 2.35. Second, dropping the three DESI DR2-input rows raises the nominal Z under both estimators; this shows influence, not that the rows are wrong or duplicate.

5 Application to a reconstruction of the 88-row table

Reconstruction. The pinned upstream repository could not be retrieved in the environment used to prepare this paper, so the 88 rows were rebuilt from Appendix A of the source paper (arXiv:2601.00650v2, Tables X and XI), with asymmetric errors reduced by the paper's own RMS rule. Of the 88 rows, 55 are exact as printed, 27 are RMS-symmetrised from two printed errors, and 6 carry one error component that was unreadable in the extracted text and is estimated (`status=approx` in `data/h0_compilation_reconstructed.csv`); the Dominguez et al. and Gonzalez et al. entries were additionally checked against their published abstracts. The reconstruction is validated against the packaged results (Table 4): the ladder group is reproduced to machine precision, and the comparison group to $0.0034\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ in the IVW mean. It is nevertheless provisional; the authoritative table is the one fetched from the pinned commit (`make data`), and every number in this section should be regenerated from it before submission.

Negative weights are explained by Corollary 1. For $(\rho_L, \rho_R) = (0.40, 0.15)$ the thresholds are $s_L^* = 2.536$ and $s_R^* = 4.218\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$: exactly the 16 ladder rows with $s_i > 2.536$ and the 38 comparison rows with $s_i > 4.218$ receive negative weight.

Feasible cross-group correlation. Across the exact PSD interval $|\rho_{LR}| \le 0.2630$ the correlation between the two pooled means is $|\kappa| \le 0.098$, and the GLS contrast varies only between $Z = 3.749$ ($\rho_{LR} = -0.263$) and $Z = 4.119$ ($+0.263$), through $Z = 3.921$ at 0. The envelope $2.80 \le Z \le 18.4$ of Proposition 2 is therefore far wider than anything attainable by a constant cross-group correlation consistent with the assumed within-group structure.

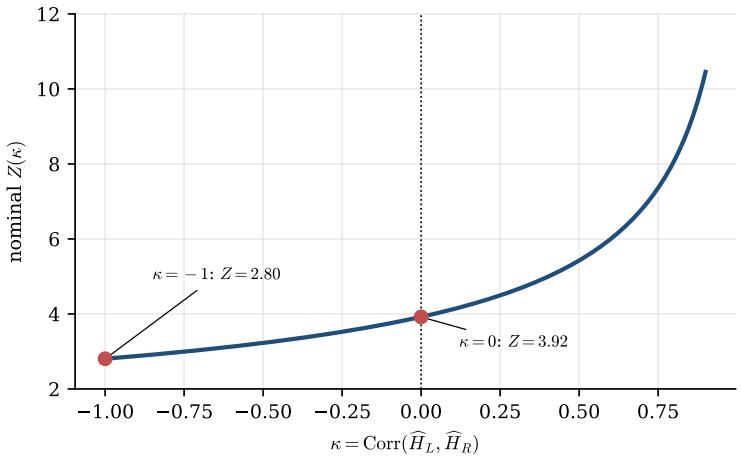


Figure 2: $Z(\kappa)$ for the GLS contrast of Table 1 (Proposition 2). Positive cross-group correlation raises Z.

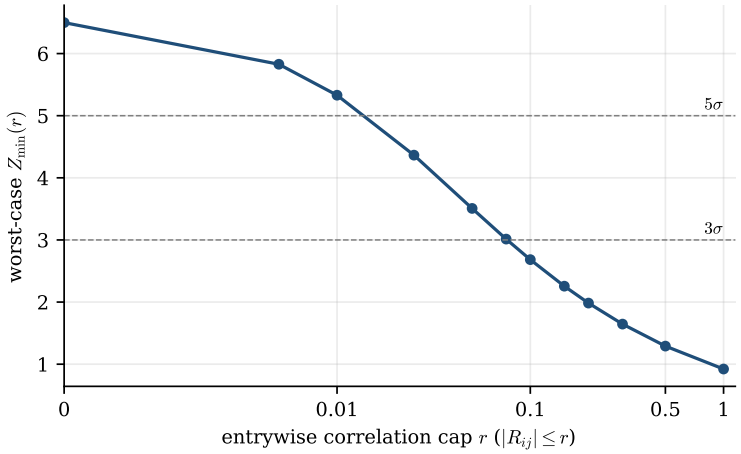


Figure 3: Worst-case $Z_{\min}(r)$ under an entrywise correlation cap (Proposition 3), for fixed inverse-variance weights.

Cap bound. The sharp bound of Proposition 3, evaluated on the reconstruction, gives $Z_{\min} =$ 6.49, 5.32, 4.36, 3.51, 2.68, 1.98, 1.29, 0.92 for $r = 0$, 0.01, 0.025, 0.05, 0.10, 0.20, 0.50, 1, matching the packaged values of Section 3.6 to better than 0.01 and equalling them at $r = 1$: an independent confirmation of both the formula and the legacy script.

Random correlation structures. Table 5 reports $Z = |\widehat{\Delta}|/\sqrt{q^\top R q}$ for the fixed inverse-variance estimator when R is drawn from LKJ(η) over all 88 rows ("joint") or independently within each group ("block"), 40 000 draws each. Consistently with Proposition 4, the distribution is narrow and centred on the independence value 6.5. The contrast between Table 5 and the equicorrelated value 3.92 is the central quantitative message: *the significance is not lost to dependence in general; it is lost to a specific assumption, namely positive correlation of equal sign across every pair in each group.* Whether such coherent shared systematics exist at the assumed strength is an empirical question that the covariance products of the primary analyses, not a prior over correlation matrices, must answer.

6 Provenance of the 69.37 ± 0.65 entry and DESI DR2 overlap

The repository row labels this estimate 2024 and cites arXiv:2405.20306 [7], whose abstract reports $69.48 \pm 0.94\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for DESI Year-1. A public presentation dated July 2025 [8]

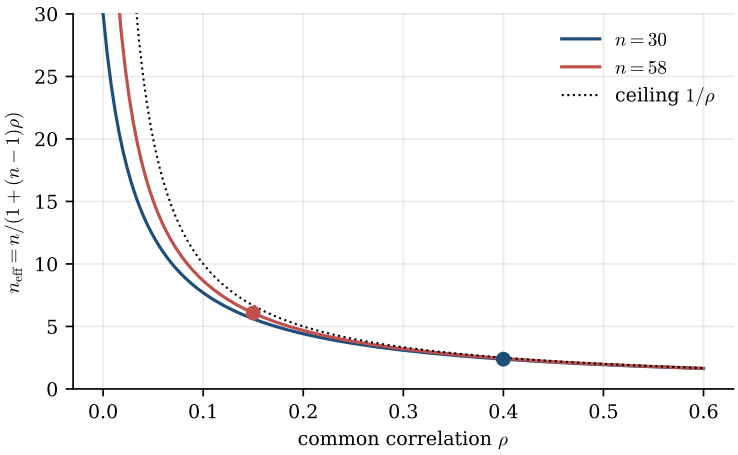


Figure 4: Effective number of independent estimates for equal errors (Proposition 1); dots mark the two assumed (n, ρ) pairs.

Table 4: Reconstruction versus packaged results (units $\mathrm{km\,s^{-1}\,Mpc^{-1}}$).

Quantity	Reconstructed	Packaged	$\lvert\Delta\rvert$
IVW ladder mean	72.732465	72.732465	0
GLS ladder mean ± SE	72.302030 ± 0.571293	72.302030 ± 0.571293	0
IVW comparison mean	69.375633	69.372226	0.0034
IVW comparison SE	0.345137	0.344887	0.0003
GLS comparison mean	69.521464	69.520526	0.0009
GLS comparison SE	0.420127	0.420384	0.0003
Negative weights (ladder / comparison)	16 / 38	16 / 38	0

reports $69.37 \pm 0.65\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ from DESI DR2 BAO, the CMB acoustic scale and a Gaussian Planck prior on ω_m, with the sound horizon free. The source paper describes the entry as a 2025 seminar/private-communication result with publication forthcoming. The matching number thus has a plausible DR2-era description, but the repository metadata does not link the row to it. *This is a traceability mismatch, not evidence of a numerical error or of misconduct*, and it is recorded as unresolved pending the authors' response.

Three comparison rows use DESI DR2 inputs: 69.37 ± 0.65, 70.03 ± 0.97 [9] and 71.59 ± 0.94 [10]. Their shares of the comparison group's diagonal inverse-variance weight are 28.15%, 12.64% and 13.46%, totalling 54.26%. That figure is a property of naive diagonal weighting. Shared DESI DR2 BAO data do not by themselves fix a pairwise covariance, which also depends on the additional likelihoods and nuisance parameters; neither independence nor perfect correlation is justified (see also [11] for a covariance-weighted treatment of local distance indicators and [12] for robust variance estimation with dependent estimates).

7 Software and reproducibility

All results of Section 3 are implemented in a header-only C++17 library with a command-line tool and mirrored in a NumPy package. Unit tests check: closed form versus dense Cholesky GLS; the negative-weight threshold; the design effect; the PSD limit (feasible just inside, infeasible just outside); the cap bound (attained, and never exceeded by random feasible matrices); and agreement of the two implementations to relative 10^{-9}.

A prior-predictive tool draws LKJ(η) correlation matrices [6] by the C-vine construction (the draw is validated against the known marginal variance $1/(2\eta - 1 + d)$) and returns the distribution of Z for fixed estimator weights; it is bit-reproducible for any thread count. It is intended for

Table 5: Prior-predictive Z under LKJ(η) correlation matrices (reconstructed table; fixed IVW weights; seed 20261026).

Structure	η	median Z	95% interval	$P(Z > 5)$
joint	1	6.52	[5.68, 7.57]	> 0.9999
joint	2	6.51	[5.69, 7.57]	> 0.9999
joint	5	6.51	[5.71, 7.52]	> 0.9999
joint	20	6.51	[5.80, 7.38]	> 0.9999
block	2	6.52	[5.65, 7.62]	> 0.9999

sensitivity exploration, not as a physical model of the dependence. Timing on a synthetic fixture with $d = 88$ and $\eta = 2$ on one core was about 0.6 ms per draw in C++ versus 89 ms in Python (about 150×). Its application to the reconstructed table is reported in Section 5; the same command reproduces it on the authoritative table after `make data`.

8 Limitations

(i) The analysis is summary-level, not a joint-likelihood refit of the primary BAO, CMB, supernova, lensing or distance-ladder data. (ii) The joint covariance of the 88 estimates is not available; ρ_L, ρ_R are the source paper's illustrative values. (iii) Cross-group covariance is set to zero except where explicitly varied. (iv) Asymmetric errors are symmetrised as upstream, discarding non-Gaussian information. (v) Exclusion scenarios keep ρ_R fixed. (vi) The provenance audit is targeted (one entry and three DESI DR2-related rows), not a row-by-row audit of all 88. (vii) Section 5 uses a provisional reconstruction of the table (six rows with an estimated error component), not the pinned upstream file. (viii) Gaussianity is assumed throughout, and the nominal Z is not a calibrated p-value under covariance uncertainty.

9 Conclusions

The reproduced inverse-variance contrast of 6.50σ falls to 3.92σ under the source paper's assumed within-group correlations, but that number is conditional on the estimator, on (ρ_L, ρ_R), and on setting the cross-group covariance to zero. The paper's mathematical contribution is to make those dependencies explicit and checkable: exact weights and their signs, the information ceiling $1/\rho$, the admissible range $|\rho_{LR}| \leq 0.263$, the monotone dependence on κ, and a sharp worst-case bound for any stated correlation cap. On the reconstructed table, generic random dependence leaves the independence significance nearly intact ($Z \in [5.7, 7.6]$), so the reduction to 3.9 rests on a specific sign-coherent correlation structure whose strength is assumed, not measured. A decisive treatment requires a row-level provenance table for all 88 entries, an overlap graph at likelihood level, and either joint covariance products or pre-specified bounds on defensible covariance structures. Until then, the tension is neither dismissed nor given a covariance-free significance.

Declarations

Data and code availability. Source data: the pinned repository snapshot of Section 2; the reconstructed table used in Section 5 is in the companion repository with per-row status flags. Code, tests, packaged tables and figure scripts: companion repository `h0-covariance-audit` (release 1.0.0; archive DOI to be added at deposit).
Competing interests. None declared. **Funding.** No external funding.
Use of AI tools. [Author to complete according to the target journal's policy.]
Scope and submission status. Technical note; to be submitted only after the checks listed in `docs/PRE_SUBMISSION_CHECKLIST.md` are complete.

References

[1] Ioannis Pantos and Leandros Perivolaropoulos. Dissecting the Hubble tension: Insights from a diverse set of sound-horizon-free H_0 measurements. *Physical Review D*, 114:063507, 2026. doi: 10.1103/9pbn-13q8.

[2] Ioannis Pantos and Leandros Perivolaropoulos. Dissecting-the-Hubble-Tension: analysis scripts for sound-horizon-independent H_0 measurements. GitHub repository, commit 5a0ddb5418a58796b39d6c1783df99ab362a9839, file `scripts/Mean_Gaussian(v2).py` (blob 9c335e27532127d4f10dc9a557de35d2aadb51cc), 2026. URL https://github.com/ipantos/Dissecting-the-Hubble-Tension/tree/5a0ddb5418a58796b39d6c1783df99ab362a9839.

[3] A. C. Aitken. On least squares and linear combination of observations. *Proceedings of the Royal Society of Edinburgh*, 55:42–48, 1935.

[4] Jack Sherman and Winifred J. Morrison. Adjustment of an inverse matrix corresponding to a change in one element of a given matrix. *Annals of Mathematical Statistics*, 21(1):124–127, 1950.

[5] Leslie Kish. *Survey Sampling*. Wiley, New York, 1965.

[6] Daniel Lewandowski, Dorota Kurowicka, and Harry Joe. Generating random correlation matrices based on vines and extended onion method. *Journal of Multivariate Analysis*, 100 (9):1989–2001, 2009. doi: 10.1016/j.jmva.2009.04.008.

[7] Levon Pogosian, Gong-Bo Zhao, and Karsten Jedamzik. A consistency test of the cosmological model at the epoch of recombination using DESI BAO and Planck measurements. 2024. URL https://arxiv.org/abs/2405.20306.

[8] Levon Pogosian. The Hubble tension and primordial magnetic fields. PASCOS 2025 presentation, slide "BAO vs CMB consistency test", 24 July 2025. URL https://conference.ippp.dur.ac.uk/event/1371/contributions/8448/attachments/6543/8955/pogosian_july24_2025.pdf.

[9] H. García Escudero, S. H. Mirpoorian, and L. Pogosian. Sound-horizon-agnostic inference of the Hubble constant and neutrino mass from BAO, CMB lensing, and galaxy weak lensing and clustering, 2025.

[10] G.-H. Du, T.-N. Li, J.-L. Ling, Y.-H. Yao, J.-F. Zhang, and X. Zhang. Model-independent late-universe measurements of H_0 and Ω_K with the parametrization based on cosmic age-improved inverse distance ladder, 2025.

[11] H0DN Collaboration. The Local Distance Network: a community consensus report on the measurement of the Hubble constant at approximately 1% precision. *Astronomy & Astrophysics*, 708:A166, 2026. doi: 10.1051/0004-6361/202557993.

[12] Larry V. Hedges, Elizabeth Tipton, and Matthew C. Johnson. Robust variance estimation in meta-regression with dependent effect size estimates. *Research Synthesis Methods*, 1(1): 39–65, 2010. doi: 10.1002/jrsm.5.